Tyler Tiede

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SUMMARY

Software engineer with 3+ years of experience in backend development, automation, and hardware-software integration. Applied Physics background with a strong problem-solving mindset. Seeking roles in backend engineering focused on building scalable Python services, REST APIs, and developer tooling.

SKILLS

Languages	Python, C++, LabVIEW
Frameworks & APIs	Flask, FastAPI, REST APIs, Click, Rich
Databases	SQLite, MySQL
Testing & CI/CD	pytest, GitHub Actions, Jenkins, Docker
Libraries	NumPy, SciPy, Pandas, Matplotlib, BeautifulSoup4
Tools & Protocols	Git, VSCode, Perforce, Serial/I²C/SCPI, Crucible
Operating Systems	Linux (Ubuntu, Raspberry Pi OS), Windows, macOS

EXPERIENCE

MKS Instruments — Rochester, NY Jan 2022 – Nov 2024

Software Engineer

- Designed and maintained Python-based regression test suites for RF plasma control systems used by major semiconductor fabs, improving release stability and confidence in firmware features.
- Built an end-to-end test platform to simulate RF generator firmware behavior and validate I²C communication with control boards, bridging hardware and software testing workflows.
- Automated data parsing and report generation pipelines, diverting developer time from manual analysis.
- Integrated serial communication with lab instruments (oscilloscopes, function generators) to enable autonomous testing of complex RF firmware features.
- Diagnosed CI pipeline test failures, improving reliability and reducing engineering support hours.

Corning, Inc. — Fairport, NY Aug – Dec 2021

Optics Technician

- Assembled customer-bound opto-mechanical systems to meet quality and performance standards.
- Collaborated with design engineers to reduce failure rates and improve manufacturing throughput.

SUNY Geneseo — Geneseo, NY Jun 2018 – Mar 2020

Assistant System Administrator

- Managed electronic lock scheduling and card access for 6,000+ users across all campus buildings.
- Assisted in developing web apps including a campus emergency alert system and internal ticketing portal using PHP, Laravel, Bootstrap, and MySQL.

PROJECTS

blamelog

Python · Click · Rich · GitPython · GitHub Actions github.com/tylerjtiede/blamelog

- Open-source Python CLI tool that layers git blame authorship with churn scoring and AST-based test gap detection to surface the highest-risk files in a codebase.
- Uses git history as a cache invalidation signal to skip redundant analysis on unchanged files, keeping runtimes fast on large repos.
- Distributed via PyPI with automated publishing via GitHub Actions CI.

lift-tracker

Python · FastAPI · SQLite · Pydantic v2 · pytest · Render github.com/tylerjtiede/lift-tracker

- REST API and web UI for logging strength training sessions, organized by named programs and reusable workout templates with per-exercise rep ranges and progression schemes.

- Implements linear progression: automatically increments target weight when rep goals are hit, with idempotent session completion so sets can be edited and re-completed without corrupting targets.
- Decoupled analysis layer computes progression trends, plateau detection, weekly adherence, and gap analysis, flagging missed muscle groups, insufficient recovery, and inactivity with rule-based suggestions.
- Full test suite covering API endpoints and analysis logic end-to-end; seed script generates 6 months of realistic training data for manual QA.

PDGA Rating Calculator

Python · Flask · BeautifulSoup4 · NumPy · SQLite · Render github.com/tylerjtiede/pdga-rating-calculator

- Reverse-engineered PDGA's rating algorithm — anchored lookback window, 24-month fallback for sparse players, outlier cutoff via std dev, and top-quartile double-weighting — validated against real historical rating updates to within ±2 points.
- Flask REST API deployed on Render powers a live web calculator; a separate CLI/GUI package adds a SQLite-backed HTTP cache, a binary-search target rating solver, and Rich terminal output.

Hardware-Software Integration Test Platform

Python · Embedded C · I²C · PyQt5 · Serial linkedin.com/in/tylerjtiede

- Designed a programmable hardware emulation platform to simulate RF generator control systems, enabling automated firmware regression testing across multiple hardware/protocol configurations.
- Developed embedded firmware for I²C command sequencing and real-time telemetry capture; built a Python GUI for manual command-and-monitor workflows during development.
- Integrated the platform into a large-scale pre-existing regression suite, extending coverage without disrupting established CI pipelines.

EDUCATION

State University of New York at Geneseo May 2021

B.S. Applied Physics · Minor in Mathematics